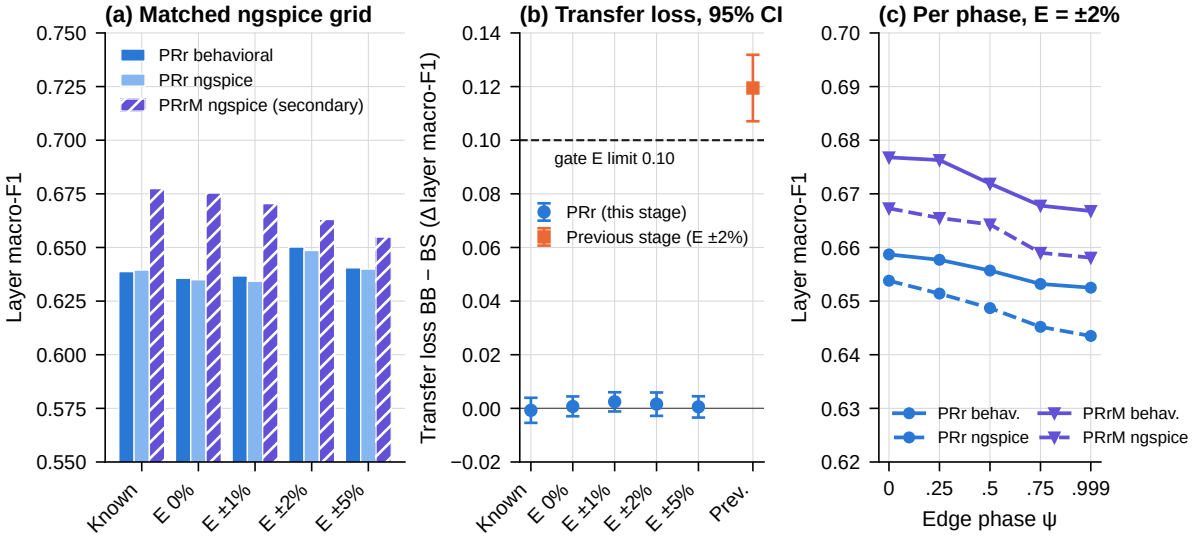
(a) Matched ngspice grid
PRr behavioral
PRr ngspice
PRrM ngspice (secondary)
Layer macro-F1
0.750
0.725
0.700
0.675
0.650
0.625
0.600
0.575
0.550
Known
E 0%
E ±1%
E ±2%
E ±5%
(b) Transfer loss, 95% CI
Transfer loss BB − BS (Δ layer macro-F1)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
gate E limit 0.10
PRr (this stage)
Previous stage (E ±2%)
Known
E 0%
E ±1%
E ±2%
E ±5%
Prev.
(c) Per phase, E = ±2%
Layer macro-F1
0.70
0.69
0.68
0.67
0.66
0.65
0.64
0.63
0.62
PRr behav.
PRr ngspice
PRrM behav.
PRrM ngspice
0
.25
.5
.75
.999
Edge phase ψ